

Tingle Tennis: Menstrual Experience Sensory Simulation Sport Device

Shun-Han Chang
National Tsing Hua University
Taiwan
shchang0204@gmail.com

Chen-Chun Wu
National Tsing Hua University
Taiwan
National Chengchi University
Taiwan
imeukg2001@gmail.com

ZI-YUN LAI
National Tsing Hua University
Taiwan
lazumeow@gmail.com

TSUNG-YEN LEE
National Tsing Hua University
Taiwan
marathon23@m112.nthu.edu.tw

CHENG-EN HO
National Tsing Hua University
Taiwan
leoho1040088@gmail.com

Min-Chun Hu
National Tsing Hua University
Taiwan
Taiwan Institute of Sports Science
Taiwan
anitahu@cs.nthu.edu.tw

Abstract

Tingle Tennis, a menstrual period sensory simulation game, leverages Virtual Reality (VR) technology, haptic feedback, and IoT techniques to create an immersive experience that highlights the physical, psychological, and social challenges athletes face during their menstrual cycles. Providing users with a situational life experience, the VR application aims to foster empathy and raise awareness about the importance of addressing menstruation-related issues in the psychology of sports and exercise. The overall experience is divided into tutorial, preparation, tennis training, and final match stages, with users assuming the role of a tennis athlete over a span of three days. The project's novelty lies in its comprehensive approach to addressing menstruation-related issues in sports by using hardware devices to mimic and control the pain of menstrual cramps based on the user's condition while experiencing the mental state of an athlete going through periods. The users also have to use various items to help users live with this situation and be fully prepared for the final match.

Keywords

virtual reality, education, pain simulation

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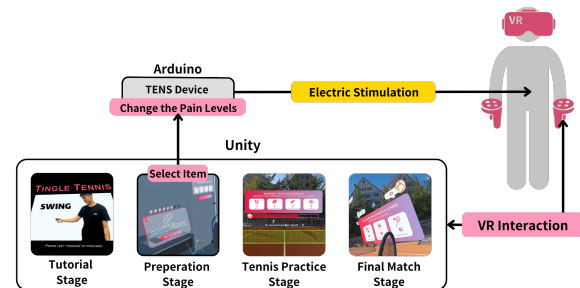


Figure 1: The four stages of the proposed menstrual experiencing system.

1 Introduction

1.1 Motivation and Previous Work

In the past, athletes who experienced irregularities due to their menstrual cycles were often forced to remain silent, as menstrual cycle was seen as an excuse. However, in recent years, the #SayPeriod movement, has aimed to encourage athletes to openly discuss their periods and the impact on their performance, as well as actively explore preparation methods to help athletes perform at their best. Our team conducted interviews with college athletes and observed that many of them faced similar challenges, like frustration and pressure from underperformance during periods and peer pressure from their teammates and coaches.

Previous work addressing menstruation-related issues primarily focused on simulating the physical aspects of menstrual symptoms. For example, Asada et al. [Asada et al. 2021] utilized an electrical stimulator to simulate the pain and discomfort associated with menstrual cramps. While this research provides valuable insights into the physical challenges during their menstrual cycles, it lacks daily, practical real-life scenarios, which are equally important aspects of a period simulation.

Table 1: The consequence for each selected item.

Item	Function
Ice coffee	Increase concentration and pain level
Hot water bag	Decrease pain level
Sanitary pad/Tampons	Increase training time
Tennis skirt	Extra training session
Pain killer	Decrease pain level(limited quantity)

1.2 Our Goals

We intend to create an immersive VR experience that highlights the day-to-day implications and discomfort of menstruation in athletes' lives by integrating a physical sensation device with elaborate VR content. As shown in Figure 1, the experiencing system is composed of four stages. In the tutorial stage, the user understands the background story and learns how to operate. Then, A preparation stage is designed for the user to prepare items for when menstruation begins. In the sports training stage, users experience varying menstrual pain levels. The final match stage incorporates gamification through a formal tennis match.

The VR simulation demonstrates menstrual pain in athletes using a nerve-stimulating device and heater. Pain levels adjust based on user choices in the preparation stage and adherence to pain management instructions during sports activities. This innovative application aims to raise awareness about menstruation in sports and contribute to gender equality discussions in athletics.

2 Design and Implementation

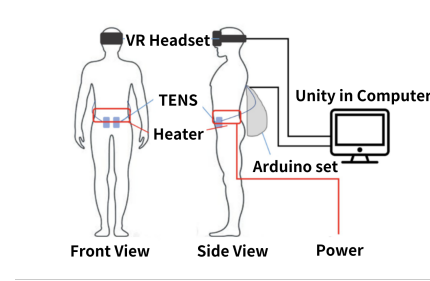
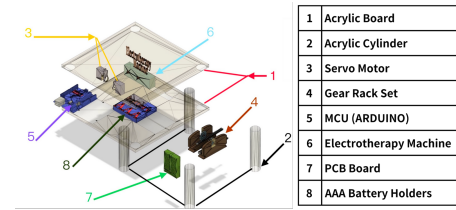
2.1 Overall Experience

As shown in Figure 1, the user first enters a *tutorial stage*, watching an introductory video that familiarizes them with the story scenario and learning instructions on how to navigate and interact. After the tutorial, users are now fully immersed in an environment as a tennis athlete. The VR content is designed to simulate a full menstrual cycle, which includes a practice stage containing six virtual days and a final match stage on the last virtual day. The preparation stage begins in the athlete's bedroom. Users interact with props to manage concentration and pain levels (Table 1 shows item-consequence mappings). Strategic decisions are crucial as items have varying effects. For instance, iced americano boosts concentration but increases pain, while painkillers offer immediate relief but have usage limits to avoid side effects.

In the tennis training stage, users aim to hit balls to increase their win probability for the final match. The system adjusts difficulty based on user proficiency, concentration, and pain levels. Users must manage menstrual flow using previously prepared sanitary products. Excessive bleeding can lead to early session termination, highlighting real challenges faced by athletes during menstruation and emphasizing preparedness.

2.2 Hardware

We developed our project via Unity and deployed it on the Quest 3 headset with controllers, taking a 3x3 meter area. To simulate the physical sensations associated with menstrual cramps, users wear a Transcutaneous Electrical Nerve Stimulator (TENS) unit and

**Figure 2: User setup diagram with the TENS device and heater equipment.****Figure 3: The detailed components of the TENS device.**

heating pads. We use Arduino uno to control these two devices (as shown in Figure 2). As users interact with the virtual environment, the Unity application sends the current pain status of the user to the computer, which then transmits the appropriate signals to the Arduino Uno. The Arduino Uno is equipped with a gear rack system that physically presses the buttons on the TENS unit, allowing it to respond to the adjustment of the stimulation intensity based on the received signals (as shown in Figure 3).

3 User Study and Future Discussion

We conducted a pilot user study from eight users. Most of the users gave positive reviews about the immersive experience and the pain simulation aspect of the project and were willing to recommend this project to people who are likely to promote related issues. However, the tutorial in game is considered insufficient, which can lead to confusion or frustration during the experience. The game scenario is also not yet able to fully simulate the uncontrollable nature of menstrual pain. The simulation for the tennis match, specifically on the racket swing, can be further refined. In the future, we will employ AI to simulate more realistic human interactions within the sports environment and refine the electrical stimulation method to more closely resemble the sensations of menstrual cramps, increase randomness, and consider more aspects of how pain affects various situations, such as during static rest. We will also continue to conduct larger-scale and more detailed user studies to further explore this topic.

References

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